

Assignment 3

Objective: Create a Battleship game that uses a 2D array to represent a game board. The player will try to "hit" hidden ships by guessing their locations.

Game Specifications:

1. Game Board Setup:

- The game board is a 8x8 grid, represented by a 2D array (`char board[8][8]`).
- Initially, all elements of the array are set to ~ (representing water).
- You will hide 7 ships on the board, represented by the letter S. Each ship occupies a single cell.
- The ships should be placed randomly on the board.

You can choose a board bigger than 8x8. The game board of 8x8 can look like this:

	1	2	3	4	5	6	7	8
1	~	~	~	~	~	~	~	~
2	~	~	~	~	~	~	~	~
3	~	~	~	~	~	~	~	~
4	~	~	~	~	~	~	~	~
5	~	~	~	~	~	~	~	~
6	~	~	~	~	~	~	~	~
7	~	~	~	~	~	~	~	~
8	~	~	~	~	~	~	~	~

2. Player's Turn:

- The player has 5 chances (or the number you choose) to guess where the ships are.
- For each guess, the player enters row and column coordinates (e.g., 2 3).
- If the guess is correct (i.e., a ship is at that location), the board should show a X at that location and print "Hit!".
- If the guess is incorrect, the board should show an O at that location and print "Miss!".
- After each guess, the game board should be displayed again with the update.

3. Ending the Game:

- The game ends when the player has used all the chances, or certain number of the ships have been hit (for example, 3 ships have been hit) .
- At the end of the game, the board is displayed with all ships revealed.

Detailed Requirements:

1. **Array Representation:** Use a 2D array to represent the board. Each element of the array represents a cell that can either be water, a hit, a miss, or a ship.
2. **Functions:** You must implement at least the following functions:
 - `void initializeBoard(char board[8][8]);`
Initializes the board with ~ (water).
 - `void placeShips(char board[8][8]);`
Randomly places 7 ships (s) on the board.
 - `void displayBoard(char board[8][8], bool revealShips = false);`
Displays the current state of the board. If `revealShips` is `true`, all the ships (s) are displayed.
 - `bool isHit(char board[8][8], int row, int col);`
Returns `true` if the player guessed the location of a ship.
 - `void playerTurn(char board[8][8]);`
Takes the player's input and marks the board accordingly (hit or miss).
3. **You can:**
 - Add more ships.
 - Allow the player to choose a custom grid size.
 - Keep track of the number of turns it took the player to win.

Grading Criteria:

- Correct implementation of game rules and logic and Proper handling of edge cases (2 points)
- Code readability, structure, and comments (1 point)
- Proper user input handling/validation and output formatting (1 point)

Submission:

Submit the C++ source file (.cpp) and the screenshot of your program output on Moodle. Test your program thoroughly and ensure that your code compiles and runs correctly.